Let us recall a few propositions from the main *Financial Maths Basics.pdf* notes:

Remark 0.1: For a self-financing strategy with $\tilde{V}_0(\phi) = 0$, if each $\tilde{S}^j_\bullet$, $j = 0, \dots, d$ were a martingale, then $\tilde{V}_\bullet(\phi)$ would be a sum of the martingale transforms of the prices by the strategy,

$$\tilde{V}_\bullet(\phi) = \sum_j (\phi^j \blacktriangleright \tilde{S}^j)_\bullet.$$

Proposition 0.1: Let $X_\bullet$ be $\mathcal{F}_\bullet$-adapted, with a time horizon N.

$X_\bullet$ is a martingale if and only if for any predictable $H_\bullet$, $\mathbb{E}[(H \blacktriangleright X)_N] = 0$.

Proposition 0.2: Any $\mathbb{R}^d$-valued predictable sequence (a "strategy" only on the risky assets)

$$\left(\phi^1_n, \dots, \phi^d_n\right) \qquad \text{for} \quad n = 0, 1, \dots$$

is a restriction of a unique self-financing strategy ($\mathbb{R}^{1+d}$-valued)

$$\left(\boldsymbol{\phi^0_n}, \phi^1_n, \dots, \phi^d_n\right) \qquad \text{for} \quad n = 0, 1, \dots$$

for any choice of an initial value $V_0(\phi) \in \mathbb{R}$ (or a choice of any of ϕ^0_n for $n \geq 0$).

Linear algebra

Remark 0.2: The set of strategies on Ω clearly forms a vector space under pointwise addition and scalar-multiplication, denote it $\mathsf{Strategy}$.

With a finite time horizon N, $\mathsf{Strategy} = \mathbb{R}^{(1+d)\times(1+N)\times\Omega}$,
and with an infinite horizon $\mathsf{Strategy} = \mathbb{R}^{(1+d)\times\mathbb{N}\times\Omega}$.

Self-financing strategies are filtered from it by the linear condition $\phi_{n+1} S_n = \phi_n S_n$, so they form a subspace $\mathsf{SelfFinancing} \subset \mathsf{Strategy}$.

Real random variables on Ω form a vector space too, denote it by $\mathbb{R}^\Omega$.

For each n and each strategy ϕ, $V_n(\phi)$ is a random variable, so a vector $V_n(\phi) \in \mathbb{R}^\Omega$. Now we can consider how the dependence on ϕ looks like:

Remark 0.3: For any n, the operator $V_n(\phi) = \phi_n \cdot S_n$ is linear in ϕ. Its signature is

$$V_n : \mathsf{Strategy} \to \mathbb{R}^\Omega$$

Then the images of $\mathsf{Strategy}$ and $\mathsf{SelfFinancing}$ are linear subspaces of $\mathbb{R}^\Omega$:

$$V_n(\mathsf{SelfFinancing}) \subset V_n(\mathsf{Strategy}) \subset \mathbb{R}^\Omega.$$

(the last inclusion might actually be an equality, TODO check)

We will talk about strategies of zero initial capital, so it is useful to have

Notation 0.4: Denote the subsets of the above strategy spaces of those strategies that satisfy $V_0(\phi) = 0$ by a superscript 0: $\mathsf{SelfFinancing}^0, \mathsf{Strategy}^0$.
Clearly they form subspaces of the respective spaces.

These terms allow us to reformulate Proposition 0.2 in a clearer way. For any $\phi = (\phi^0, \phi^1, ..., \phi^d) \in \mathsf{Strategy}$, denote by $\underline{\phi}$ the risky part of ϕ, $\underline{\phi} := (\phi^1, ..., \phi^d)$. Then the projection

$$\begin{aligned} \mathsf{Strategy}_{1+d} &\to \mathsf{Strategy}_d \\ \phi &\mapsto \underline{\phi} \end{aligned}$$

is a linear map, where the d and $d + 1$ subscripts denote the number of assets the space considers.

Remark 0.5: Proposition 0.2 establishes that the linear map

$$\begin{aligned} \mathsf{SelfFinancing}_{1+d} &\cong \mathsf{Strategy}_d \times \mathbb{R} \\ \phi &\mapsto \left(\underline{\phi}, V_0(\phi)\right) \end{aligned}$$

is in fact an isomorphism.

Remark 0.6: This allows us to compute $\dim(\mathsf{SelfFinancing}_d) = \dim\left(\mathbb{R}^{d\times(1+N)\times\Omega} \times \mathbb{R}\right) = 1 + d \times (1 + N) \times \Omega$ and so

$$\begin{aligned} \operatorname{codim}(\mathsf{SelfFinancing}_{1+d}) &= \dim(\mathsf{Strategy}_d) \qquad - \dim(\mathsf{SelfFinancing}_{1+d}) \\ &= (1 + d) \times (1 + N) \times \Omega - (d \times (1 + N) \times \Omega + 1) \\ &= (1 + N) \times \Omega - 1. \end{aligned}$$

Arbitrage

Definition 0.7 (Admissible strategy): A *self-financing* strategy is called **admissible** if $V_n(\phi) \geq 0$ for all n (and all $\omega \in \Omega$).

Remark 0.8: The set of admissible strategies is closed under addition and non-negative scalar multiplication, so it forms a cone $\mathsf{Admissible} \subset \mathsf{SelfFinancing}$.

Its image under V_n is then a cone $V_n(\mathsf{Admissible}) \subset V_n(\mathsf{SelfFinancing}) \cap \mathbb{R}^{\Omega}_{+}$.

Like above, denote the set of zero initial capital admissible strategies by $\mathsf{Admissible}^0$.

Thus we get the following chain of strategy classes (here N is either a finite number or equals $\mathbb{N}$, and $\leq$ means $\subset$ but for subspaces):

$$
\mathbb{R}^{(1+N)\times(1+d)\times\Omega}_{\geq 0}
\begin{array}{c} \supset \\ \\ \supset \end{array}
\begin{array}{ccccc}
\mathsf{Admissible} & \leq & \mathsf{SelfFinancing} & \leq & \mathsf{Strategy} \\
\cup & & \rotatebox{90}{$\leq$} & & \rotatebox{90}{$\leq$} \\
\mathsf{Admissible}^0 & \leq & \mathsf{SelfFinancing}^0 & \leq & \mathsf{Strategy}^0
\end{array}
\begin{array}{c} \lneq \\ \\ \lneq \end{array}
\mathbb{R}^{(1+N)\times(1+d)\times\Omega},
$$

and for each $n \leq N$, the corresponding chain of images under $V_n : \mathbb{R}^{N\times\Omega} \to \mathbb{R}^{\Omega}$:

$$
\mathbb{R}^{\Omega}_{\geq 0}
\begin{array}{c} \supset \\ \\ \supset \end{array}
\begin{array}{ccccc}
V_n(\mathsf{Admissible}) & \leq & V_n(\mathsf{SelfFinancing}) & \leq & V_n(\mathsf{Strategy}) \\
\cup & & \rotatebox{-90}{$\leq$} & & \rotatebox{90}{$\leq$} \\
V_n(\mathsf{Admissible}^0) & \leq & V_n(\mathsf{SelfFinancing}^0) & \leq & V_n(\mathsf{Strategy}^0)
\end{array}
\begin{array}{c} \lneq \\ \\ \lneq \end{array}
\mathbb{R}^{\Omega}.
$$

We fix an $N \in \mathbb{N}$, called the *horizon*. All indices n, i, j, k below vary between 0 and N.

The market is said to have an **arbitrage opportunity** if some admissible strategy with zero initial value delivers a strictly positive value on a non-null set.
That is, an arbitrage opportunity is a

zero-investment	**risk-free**	**chance of profit**.
($V_0 = 0$)	($V_n \geq 0$ for *every* ω and n)	($V_N \gneqq 0$ on a non-null set)

A market without arbitrage opportunities is called viable.

Definition 0.9 (Viable/arbitrage-free market): The market is called **arbitrage-free** or **viable** if every admissible strategy with $V_0(\phi) = 0$ satisfies $V_N(\phi) = 0$.

In terms of the notation above, a market is viable if $V_N(\mathsf{Admissible}^0) = 0$.

Spelled out fully, the market is viable if for all self-financing ϕ,

$$
\text{if} \quad \begin{array}{c} \forall n : V_n(\phi) \geq 0 \\ \text{and } V_0(\phi) = 0 \end{array}, \text{ then } V_N(\phi) = 0.
$$

As a silly (counter)example, if

$$
\begin{aligned}
&\text{at} \quad n = 0, \quad S^1_0 = S^0_0 \\
&\text{at} \quad n > 0, \quad S^1_n = 2 \cdot S^0_n,
\end{aligned}
$$

i.e. the price of risky asset 1 equals the riskless at first but then becomes twice as much as the riskless one, then the constant strategy $\phi_n = (-1, 1)$ has zero initial value and $V_n = -S^0_n + S^1_n = S^0_n > 0$. Thus the market defined by those $S^{\bullet}_{\bullet}$ is not viable.

A viable market was defined by a condition on the terminal portfolio value $V_N(\phi) = 0$ for all *admissible* strategies, but it is equivalent to require it from all self-financing strategies:

Proposition 0.3: The risk-freeness condition ($V_n(\phi) \geq 0$) for arbitrage-free markets can also be required only for horizon time N:

A market is viable	if and only if	no strategy $\phi \in \mathsf{SelfFinancing}^0$ can provide $V_N(\phi) \in \mathbb{R}^{\Omega}_{\geq 0} \setminus \{0\}$.

Proof: The ($\Leftarrow$) direction is trivial, just because $\mathsf{Admissible}^0 \subset \mathsf{SelfFinancing}^0$.

For ($\Rightarrow$), assume that no $\phi \in \mathsf{Admissible}^0$ gives $V_N(\phi) \in \mathbb{R}^{\Omega}_{\geq 0} \setminus \{0\}$ and take a $\phi \in \mathsf{SelfFinancing}^0$ to show $V_N(\phi) \notin \mathbb{R}^{\Omega}_{\geq 0} \setminus \{0\}$.

If $\phi \in \mathsf{Admissible}^0$, we're done; now assume ϕ is not admissible, i.e. gives a negative portfolio value $V_n(\phi) < 0$ with positive probability at some time n.

If $n = N$, we're done; now take n to be the last such one. Then the strategy ϕ with positive probability has negative value at $n < N$ and at all later times gives $V_\bullet(\phi) \in \mathbb{R}^{\Omega}_{\geq 0}$.

TODO □

Fundamental theorem of asset pricing

Proposition 0.4 (Fundamental theorem of asset pricing):

A market is arbitrage-free	if and only if	under some equivalent measure, the discounted price processes $\tilde{S}^k_\bullet, k = 0, 1, \ldots$ are martingales.

Before we present the proof, this theorem deserves a bit of discussion.

Remark 0.10: Now the L&L book proceeds to prove this directly. I find the presentation of the proof there far from pedagogical, because they combined 4 distinct concepts in the same proof without clear indication which one is used where and how they are connected:
1. Interpreting viability in terms of only at-the-end lack of risk;
2. Treating martingales as processes having all their increments-transforms of zero-mean;
3. Relationship between value of self-financing strategies and martingale transforms;
4. Separation of a convex set from a linear subspace;

They use all (1),(2),(4) only in the ($\Rightarrow$) direction, and (3) only for ($\Leftarrow$), and to me that makes the easy direction's (the one that does not use convex separation) proof look mysterious. But if we notice that (1) and (2) concern exclusively the left and right side of the equivalence, respectively, and (3) applies to each side individually, we can apply them upfront and make the proof conceptually shorter.

(1) and (2) were already discussed as propositions above, so we can readily install them in the two sides of the theorem statement.

Applying the equivalence from Proposition 0.3 and the characterization of martingales in Proposition 0.1, the proposition reads

For any $\phi \in \mathsf{SelfFinancing}^0$, if $V_N(\phi) \geq 0$, then $V_N(\phi) = 0$.	if and only if	there exists $\mathbb{P}^*$ equivalent to $\mathbb{P}$, such that for any predictable $H_\bullet$, $\mathbb{E}^*\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0 \quad \text{for all } j.$

Remark 0.11: I find this phrasing illuminating, because by Remark 0.1 we know that self-financing strategies of zero initial value, when applied on martingale prices, act like martingale transforms. And here on the left side we have exactly such strategies, and on the right side - martingale transforms of the prices.

In fact, can we rewrite the right side in terms of portfolio values of zero-investment self-financing strategies then? That is, can we say that

Proposition 0.5: The way predictable $\mathbb{R}$-valued sequences act on process increments and the portfolio value of self-financing strategies with zero initial capital are related by the equivalence

For all predictable $\overset{(\mathbb{R}\text{-valued})}{H}$ and each $\overset{(\text{asset})}{j}$, $\mathbb{E}\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0$	if and only if	for all $\phi \in \mathsf{SelfFinancing}^0$, $\mathbb{E}\left[\tilde{V}_N(\phi)\right] = 0$

(notice the similarity between the first lines on each side as well as between the second lines)

Proof: As discussed in Remark 0.1, for a self-financing strategy, the discounted portfolio value is a martingale transform of the market prices, $\tilde{V}_N(\phi) = \sum_{j=0}^{d}\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N$, so by linearity of $\mathbb{E}$ the right side condition can be replaced by $\sum_j \mathbb{E}\left[\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N\right] = 0$. We prove the rest in the next proposition. □

Proposition 0.6:

For all predictable $\overset{(\mathbb{R}\text{-valued})}{H}$ and each $\overset{(\text{asset})}{j}$, $\mathbb{E}\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0$	if and only if	for all $\phi \in \mathsf{SelfFinancing}^0$, $\sum_j \mathbb{E}\left[\phi^j \blacktriangleright \tilde{S}^j\right] = 0$

Proof ($\Rightarrow$): Assume the left side and take a $\phi \in \mathsf{SelfFinancing}^0$. All ϕ^j are surely predictable, so each $\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N$ has zero mean and the total gain expectation vanishes:

$$\mathbb{E}^*\left[\tilde{V}_N\right] = \mathbb{E}^*\left[\sum_j \left(\phi^j \blacktriangleright \tilde{S}^j\right)_N\right] = \sum_j \underbrace{\mathbb{E}^*\left[\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N\right]}_{0} = 0.$$

□

Before showing the other direction, a brief comment:

Remark 0.12: Notice that $\tilde{S}^0$ is constant, so trivially $H \blacktriangleright \tilde{S}^0 = 0$ and the condition on the left side here matters only for $j > 0$. This is why in the proof in the book, instead of self-financing strategies, they consider predictable processes in $\mathbb{R}^d$ (and not $\mathbb{R}^{d+1}$). But I find this unnecessarily confusing, as we can reuse terms already established ($\mathsf{Admissible}^0$) and just point out that the j on the right hand side can vary from 1 onwards.

Proof ($\Leftarrow$): Take a predictable H and an asset j, we have to prove that $\mathbb{E}^*\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0$.

If $j = 0$, we are done by the preceding remark, so now H can be regarded as a strategy operating only on a risky asset $j > 0$.
By Proposition 0.2 we can extend H to a (unique) self-financing strategy ϕ with $V_0(\phi) = 0$.

But for $\phi \in \mathsf{SelfFinancing}^0$ we've assumed

$$\sum_j \mathbb{E}\left[\phi^j \blacktriangleright \tilde{S}^j\right] = 0.$$

All sum terms except $i = j$ vanish because for $i = 0$, $\tilde{S}^i$ is constant; while for $i > 0$ and $i \neq j$, ϕ^j is zero. And ϕ^j is just H, so the sum consists simply of the term $\mathbb{E}\left(H \blacktriangleright S^j\right)_N$ and it is zero.□

Now we can rephrase the right side of the fundamental theorem of asset pricing again:

For any $\phi \in \mathsf{SelfFinancing}^0$, if $V_N(\phi) \geq 0$, then $V_N(\phi) = 0$.	if and only if	there exists $\mathbb{P}^*$ equivalent to $\mathbb{P}$, s.t. for any $\phi \in \mathsf{SelfFinancing}^0$, $\mathbb{E}^*\left[\tilde{V}_N(\phi)\right] = 0.$

Assuming that $\mathcal{F} = \mathcal{P}(\Omega)$ and $\mathbb{P}(\omega) > 0$ for all outcomes $\omega \in \Omega$, the equivalence $\mathbb{P}^* \sim \mathbb{P}$ means $\forall \omega : \mathbb{P}^*(\omega) > 0$ as well. A mere geometrical observation should now make the statement almost obvious:

Remark 0.13: Considering the object $\mathbb{P}^*$ as a vector in $\mathbb{R}^\Omega$, for any ϕ the mean $\mathbb{E}^*\left[\tilde{V}_N(\phi)\right]$ is just the dot product $\mathbb{P}^* \cdot \tilde{V}_N(\phi)$. Then the condition $\mathbb{E}^*\left[\tilde{V}_N(\phi)\right] = 0$ for all $\phi \in \mathsf{SelfFinancing}^0$ means just that $\mathbb{P}^*$ must be orthogonal to $\tilde{V}_N(\mathsf{SelfFinancing}^0)$.

So the theorem statement can be equivalently phrased as

Proposition 0.7 (Fundamental theorem of asset pricing, again):

$$V_N(\mathsf{SelfFinancing}^0) \cap \mathbb{R}^\Omega_{\geq 0} = \{0\} \quad \text{if and only if} \quad \exists \mathbb{P}^* \in \mathbb{R}^\Omega_{\geq 0} : \mathbb{P}^* \perp \tilde{V}_N(\mathsf{SelfFinancing}^0)$$

Now *that* phrasing should be geometrically quite clear.

Proof:

($\Leftarrow$). Assume $\mathbb{P}^*$ equivalent to $\mathbb{P}$ such that $\mathbb{P}^* \perp \tilde{V}_N(\mathsf{SelfFinancing}^0).$	($\Rightarrow$). Assume $\tilde{V}_N(\mathsf{SelfFinancing}^0) \cap \mathbb{R}^\Omega_{\geq 0} = \{0\}.$

Take an element

$$f \in \underbrace{V_N(\mathsf{SelfFinancing}^0)} \cap \underbrace{\mathbb{R}^\Omega_{\geq 0}}$$

$$\text{Then} \qquad \mathbb{P}^* \perp f \quad \text{and} \quad f \geq 0,$$

i.e. $\mathbb{E}^*(f) = 0$ and $f \geq 0$, so $f \overset{\mathbb{P}^*}{=} 0$.

The left term is a linear space and the right term is convex, so by a standard separation theorem there exists a separating functional $\mathbb{E}^*$ on $\mathbb{R}^\Omega$ vanishing on $\tilde{V}_N(\mathsf{SelfFinancing}^0)$ and positive on $\mathbb{R}^\Omega_{\geq 0}$, or — a unit (by $\|\cdot\|_1$) vector $\mathbb{P}^* \in \mathbb{R}^\Omega_{\geq 0}$ orthogonal to $\tilde{V}_N(\mathsf{SelfFinancing}^0)$.

□

Summary

We summarize the proof and the preceding rewrites of the Fundamental theorem of asset pricing (FTAP).

Proposition 0.8 (Fundamental theorem of asset pricing): For a market defined by the $\mathbb{R}^{1+d}$-valued process $S_\bullet$, the following equivalences hold:

$$\begin{tikzcd}
\begin{array}{c}\text{The market is}\\ \text{arbitrage-free}\end{array} \arrow[r, Leftrightarrow, "\text{FTAP}"] \arrow[dd, Leftrightarrow, "\text{By definition}"'] & \begin{array}{c}\text{Under some } \mathbb{P}^* \text{ equivalent to } \mathbb{P},\\ \text{all } \tilde{S}^j_\bullet \text{ are martingales.}\end{array} \arrow[d, Leftrightarrow, "\text{Proposition 0.1}"] \\
& \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall H \text{ predictable } \forall j\\ \mathbb{E}^*\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0\end{array} \arrow[d, Leftrightarrow, "\text{Proposition 0.5}"] \\
\begin{array}{c}\forall \phi \in \mathsf{Admissible}^0\\ V_N(\phi) = 0\end{array} \arrow[dd, Leftrightarrow, "\text{Proposition 0.3}"'] & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall \phi \in \mathsf{SelfFinancing}^0 :\\ \sum_j \mathbb{E}^*\left[\phi^j \blacktriangleright \tilde{S}^j\right] = 0\end{array} \arrow[d, Leftrightarrow, "\text{Proposition 0.6}"] \\
& \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall \phi \in \mathsf{SelfFinancing}^0\\ \mathbb{E}^*[V_N(\phi)] = 0\end{array} \arrow[d, Leftrightarrow, "\text{Remark 0.13}"] \\
\begin{array}{c}\forall \phi \in \mathsf{SelfFinancing}^0\\ V_N(\phi) \geq 0 \Rightarrow V_N = 0\end{array} \arrow[d, Leftrightarrow, "\text{Syntactical}"'] & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall \phi \in \mathsf{SelfFinancing}^0\\ \mathbb{P}^* \cdot V_N(\phi) = 0\end{array} \arrow[d, Leftrightarrow, "\text{Syntactical}"] \\
\begin{array}{c}V_N(\mathsf{SelfFinancing}^0) \cap \mathbb{R}^\Omega_{\geq 0}\\ = \{0\}\end{array} \arrow[r, Leftarrow, shift left, "\text{Hahn-Banach}"] \arrow[r, Rightarrow, shift right, "\text{Measure-theoretic}"'] & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} :\\ \mathbb{P}^* \perp V_N(\mathsf{SelfFinancing}^0)\end{array}
\end{tikzcd}$$

Example for $|\Omega| = 2$

The Fundamental theorem of asset pricing is essentially a statement about the space $\mathbb{R}^\Omega$, in which portfolio terminal values lie. $\mathbb{R}^\Omega$ is the space of all possible random choices for a random real number.

That space is $|\Omega|$-dimensional, so we can visualise graphically it when the probability space is as simple as $\Omega = \{\omega_1, \omega_2\}$, i.e. there are only two possible outcomes.

An arbitrage strategy will have its portfolio value lie in the first quadrant of $\mathbb{R}^{\{\omega_1,\omega_2\}}$ (that is, $\{(x,y) : x \geq 0, y \geq 0\}$), so that in no case of the two outcomes it induces a loss.

Equivalently, a no-arbitrage self-financing strategy of zero-investment will have its final portfolio value *outside* of that quadrant, so for a viable market all terminal values of $\mathsf{SelfFinancing}^0$ form a line through the origin that does not enter the first quadrant. Then that line induces a unit normal lying in that quadrant so gives positive probabilities to both outcomes (i.e. is an equivalent probability measure).